Soh Jin Xin Shana

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Final-year Computer Science & Design undergraduate at SUTD (graduating by June 2027) who bridges technical implementation and business outcomes, with hands-on experience in requirements gathering, full-stack development and cloud infrastructure across government and research platforms. Delivered features for a platform used by 2,290+ built environment professionals.

Education

Singapore University of Technology & Design — **Sep 2023 – May 2027**
Bachelor of Engineering in Computer Science and Design (CGPA: 3.96/5.0) — *Singapore*
- Relevant Modules: Software Engineering, Database Systems, Algorithms, Design Thinking

University of Newcastle — **Jan 2026 – May 2026**
Global Exchange Program (2 High Distinctions, 2 Distinctions) — *Newcastle, Australia*
- Courses: Software Verification & Validation, Cybersecurity, Games Design, Sociology

Nanyang Polytechnic — **Apr 2020 – Apr 2023**
Diploma of Business and Financial Technology (CGPA: 3.47/4.0) | Director's List — *Singapore*

Experience

Software & Product Engineer Intern — **May 2026 – Present**
City Syntax Lab, National University of Singapore (NUS) — *Singapore*
- Act as requirements contact for government agencies, researchers and testers on the Singapore Building Carbon Calculator, translating feedback into prioritized, implementable features through agile iterative delivery
- Integrated Stripe payments end-to-end (NestJS), choosing embedded in-app checkout over a hosted redirect to keep users in-application; secured fulfilment with signature-verified webhooks and session-level idempotency for exactly-once credit delivery
- Designed a credit-wallet pricing model with margin analysis against third-party inference (Stability AI) API costs, tying platform pricing to underlying compute spend
- Migrated staging and production databases from Supabase to AWS RDS (PostgreSQL 15) using Terraform workspaces for reproducible infrastructure-as-code; implemented application-wide API rate limiting (NestJS Throttler)
- Initiated the team's project planning documentation covering scope, milestones and action plans, improving cross-team alignment and delivery clarity

Software Engineer Intern — **Sep 2025 – Dec 2025**
Building and Construction Authority (BCA), Digitalisation Team — *Singapore*
- Key contributor to a full-stack web app (Node.js, TypeScript, TypeORM, SQL) used by 2,290+ built environment professionals, supporting ∼58 active projects and ∼1,460 site visits
- Ran user interviews and UAT, analyzing feedback to prioritize product enhancements, and presented technical milestones to department leadership
- Automated 100% of a department's manual deadline tracking via a time-based email notification system (PlumberSG); productized Python scripts into one-click executables (Tkinter) for non-technical teams
- Translated complex IFC Calculator logic into flowcharts and documentation aligning engineering and business users; built an export feature mapping carbon emissions data to industry-standard Excel templates

Multimedia Experiential Developer Intern — **May 2024 – Jul 2024**
Zing Experiential Studio — *Singapore*
- Built a high-fidelity Next.js prototype for a gamified educational platform (TypeScript, Tailwind CSS), owning core UI and user-flow logic for an experience that engaged 70,000+ visitors
- Developed an animated touchscreen interface (Unity, C#, LeanTween) deployed at the World Cities Summit, engaging 1,000+ delegates with ESG frameworks; facilitated client meetings to present milestones and iterate on feedback

Academic & Technical Projects

DiceRealm (SUTD CSD Showcase) | *Java, Spring Boot, Android, OpenAI API* — **Feb 2025 – Apr 2025**
- Led front-end product development in a 7-person agile team, defining core gameplay loops and implementing MVVM architecture with concurrent game logic for real-time AI-narrated dialog and combat

Thrive-Thru Interactive Showcase (SUTD DTI) | *Design Thinking* — **Feb 2024 – Apr 2024**
- Applied Design Thinking in a 5-person team to prototype an interactive display for an institutional branding problem, designing a track-guided journey with synchronized lighting that kept the interaction under one minute

Leadership & Extra-curricular Experience

Youth Camp Committee, Publicity I/C — **Mar 2022 – Dec 2022**
- Led a cross-functional team over a 9-month planning cycle, delivering digital publicity assets, branding, the official logo, registration and end-to-end event logistics

Harvard Business Club Mentorship Programme — **2021 – 2022**
- Delivered a keynote on 'Personal Resilience' to 200+ members, synthesizing takeaways on diverse workplace cultures and professional development

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C#, SQL, HTML, CSS
Frameworks: Next.js, React, Node.js, NestJS, Spring Boot, Tailwind CSS, Unity
Cloud & Infrastructure: AWS (RDS, ECS, EC2), Terraform, PostgreSQL, Supabase, TypeORM, REST APIs, Stripe API
Product & Process: Requirements gathering, stakeholder management, user interviews, UAT, agile iteration, prototyping, Figma, Git